A 192-Gb 12-High 896-GB/s HBM3 DRAM With a TSV Auto-Calibration Scheme and Machine-Learning-Based Layout Optimization

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***Abstract*— This article introduces a 192-Gb 896-GB/s 12-high stacked third-generation high-bandwidth memory (HBM3 DRAM) with low power consumption and high-reliability traits. New design schemes and features, including internal low-voltage signaling, center strobe calibration, through-silicon via (TSV) auto-calibration, a symbol-correcting in-DRAM ECC, and machine-learning-based layout optimization, allow large amounts of data transfers among the vertically stacked base and core dies with limited delay mismatch or SI degradation, as well as reduced power consumption from low-voltage swings. Experimental results confirm 896-GB/s bandwidth operations at 1.0-V voltage conditions with up to 15% improved power efficiency.**

***Index Terms*— 3-D-stacked memory, decision-feedback equalizer (DFE), high-bandwidth memory (HBM), in-DRAM ECC, machine learning, through-silicon via (TSV).**

I. Introduction

With the advent of the world's first artificial intelligence (AI) model perceptron [1] in 1960, the progress of compute performance growth has been traditionally consistent with AI model advancements. This assertion is evidenced by the correlation between daily compute training performance and AI advancement trends [2] that confirm training performance trends that double every 24 months in line with Moore's law [3] until early 2010. This is in contrast to the follow-up 3.4-month cycle starting from 2012 when training performance begins to outpace conventional macro-level advancements. Future AI models with capacities of hundreds of billions of parameters and state-of-the-art workload and architecture are expected to further fuel compute training performance to surpass beyond existing boundaries.

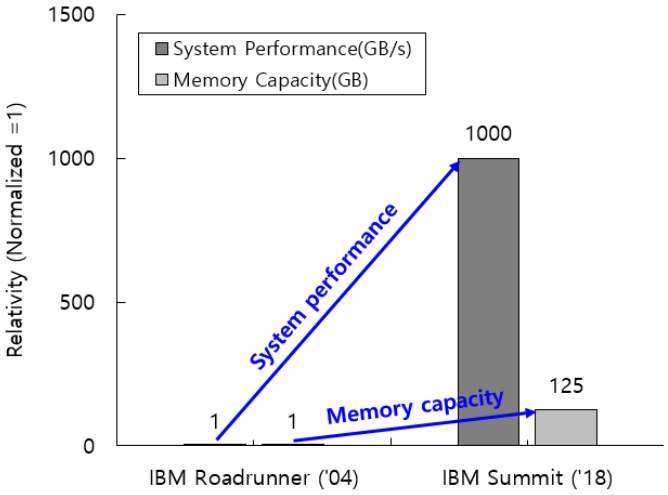


Fig. 1. System performance and memory capacity comparison of supercomputers.

From the real-life application perspective, this trend is confirmed by the rapidly increasing level of performance growth by supercomputers generally positioned at the pinnacle of the system-level performance hierarchy. Fig. 1 shows a comparison of system performance and memory capacity between two representational supercomputers that officially broke the petaFLOP/s and exaFLOP/s performance boundaries in the early 2000s and the late 2010s, respectively. It is notable that, despite the thousandfold increase in the system-level performance during the defined decade, the increase rate of memory was limited to 125x over the same course of time. Even with taking different factors into account, the two

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TABLE I
SUMMARY OF HBM DRAM PRIOR ARTS

	HBM1 [5]	HBM2 [6]–[8]	HBM2E [9], [10]
Pin Data Rate	1.0Gbps	2.0Gbps ∼2.67Gbps	4.0Gbps ∼5.0Gbps
Max Bandwidth	128GB/s	256GB/s ∼341GB/s	512GB/s ∼640GB/s
Density	1GB	8GB	16GB
Burst Length	2/4	2/4	4
Channel	8CH	8CH, 16p-CH	8CH, 16p-CH
Supply Voltage	VDDC/Q=1.2V VPPE=2.5V	VDDC/Q=1.2V VPPE=2.5V	VDDC/Q=1.2V VPPE=2.5V
Die Dimension	<8mm×12mm	8mm×12mm	10mm×11mm

contrasting growth rates imply that there is a striking level of imbalance between real-life system requirements versus memory offerings.

Most recently, high-bandwidth memory (HBM) DRAM has been heralded as a prominent solution to tackle the above-mentioned growing disparity between processor and memory bandwidth known as the "memory wall" [4]. Utilizing 3-D vertical die stack technology based on multiple DRAM core and base logic die, optimization of sub-independent pseudo-channel, and bank group organizations, as well as the extension of x1024 IOs, HBM DRAM stretches the existing memory bounds set by conventional DRAM by offering the highest level of memory bandwidth, density, power efficiency, and form factor. Table I presents an overview of former HBM DRAM generations, e.g., HBM1 [5], HBM2 [6]–[8], and HBM2E [9], [10], which are widely adopted by key system-level platforms. Despite notable memory performance advancements, however, the competitive rush of high-tier system launches, including supercomputers, hyper-scale data centers, and machine learning accelerators, is further expediting requirements for higher bandwidth memory solutions.

In this article, we introduce the world's first third-generation HBM (HBM3) DRAM that supports multiple new features and design schemes to accommodate the increasing system-level demands. Section II illustrates the proposed high-level architecture and key components of HBM3 DRAM. Section III explains the internal low-voltage signaling scheme utilized for low-power designs to prevent thermal constraints. Section IV introduces the center strobe calibration scheme that automatically adjusts the optimal timing across PVT variations. Section V covers the machine-learning-based design layout optimized to reduce group signal skew effects at the internal I/O bus during read and write operations to align DQ signals for proper signaling. Section VI presents the through-silicon via (TSV) calibration feature that minimizes TSV interface tradeoffs between multiple 3-D stack ranks at high data transfer rates. Section VII describes the on-die (OD) ECC scheme based on the Reed–Solomon code used to mitigate DRAM malfunction issues that commonly originate from on-DRAM cell failures. Last but not least, Section VIII summarizes the experimental results of the proposed HBM3 DRAM that confirms the world's fastest DRAM bandwidth performance at low supply voltage conditions formulated on a representation of system-level interposer package and test methodology, and Section IX concludes this article.

II. OVERALL ARCHITECTURE

Fig. 2 shows the stacked diagram of the proposed HBM3 DRAM. HBM3 DRAM supports higher density and improved parallelism supporting up to 12-high die stack and 16 independent channels, in comparison to the previous HBM2E DRAM that was limited to eight-high die stack and eight independent channels [9], [10]. The proposed architecture includes four channels per core die with four slices forming a single unit rank, or a total of 12 slices corresponding to three ranks.

The core dies are vertically stacked on a base die and interconnected via TSVs, where the TSV locations are adjusted at the base die. As the base die concurrently exploits advanced technology and additional metal layers, the TSV location adjustment is more relevant at the base die than at the core die. That is, if the adjustment were done in the core die, the core die needs more metal layers and increases the manufacturing cost as an HBM stack consists of a base die and multiple core dies. Following the adjustment, each channel unit is identically designed, as shown in Fig. 2, enhancing the optimization and management of each channel unit design. A unit channel is subdivided into two pseudo-channels where each pseudo-channel consists of four bank-grouped 16 banks with 160-Mb density, including 16 Mb of parity cells for ECC and 8 Mb of metadata cells for a system-level reliability check. Each bank group has 272 dedicated I/O (BK IO) lines, and four sets of BK IOs share two sets of global lines located in the TSV area with time-division multiplexing to secure a sufficient timing window of 2 tCK.

In addition to the benefits of cost and identical design, TSV-to-core die alignment has advantages in the aspect of thermal dissipation. As core dies are bigger than the PHY area, the TSV area is larger for TSV-to-core die alignment compared to the TSV-to-PHY alignment. In this design, the additional TSV area is about 40%, and this larger TSV area helps to reduce the overall thermal resistance of the stacked DRAMs by lowering the amount of power consumed per unit area. Thermal analysis results with the worst case scenario shown in Fig. 3 confirm that the TSV-to-core die alignment results in a lower junction temperature of 101.4 °C compared to 102.7 °C of the TSV-to-PHY alignment case and satisfies the thermal criterion of 105 °C.

Nevertheless, adjusting thousands of lines without delay mismatch or SI degradation in a single base die is technically challenging and consumes large amounts of power.

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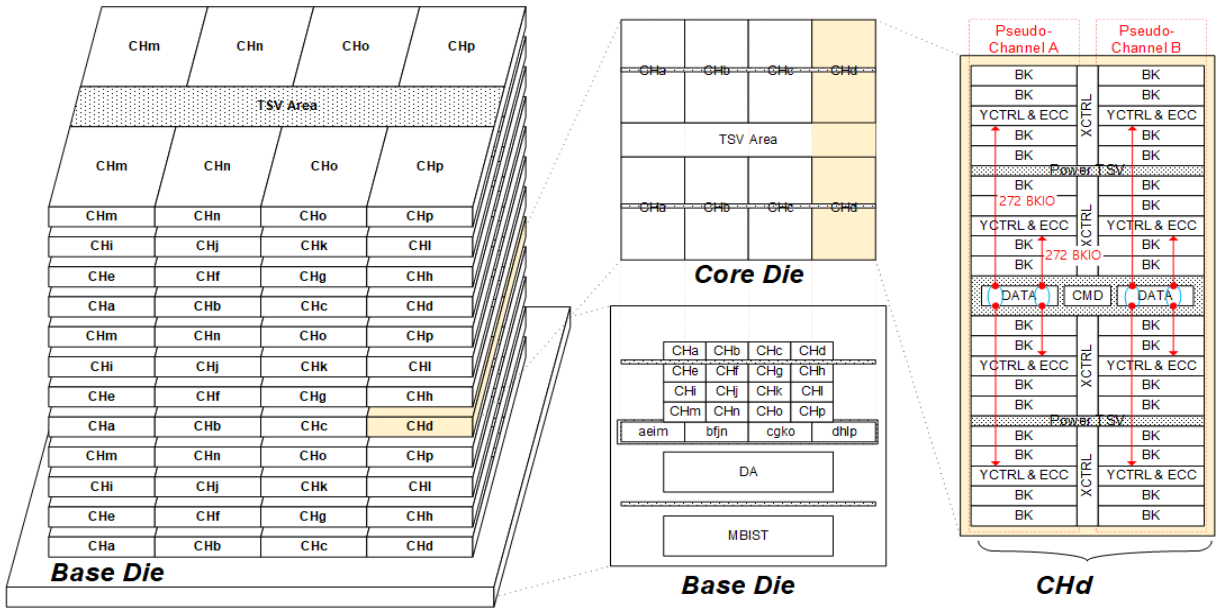


Fig. 2. Stack view of 12-high HBM and the block diagram of the unit channel.

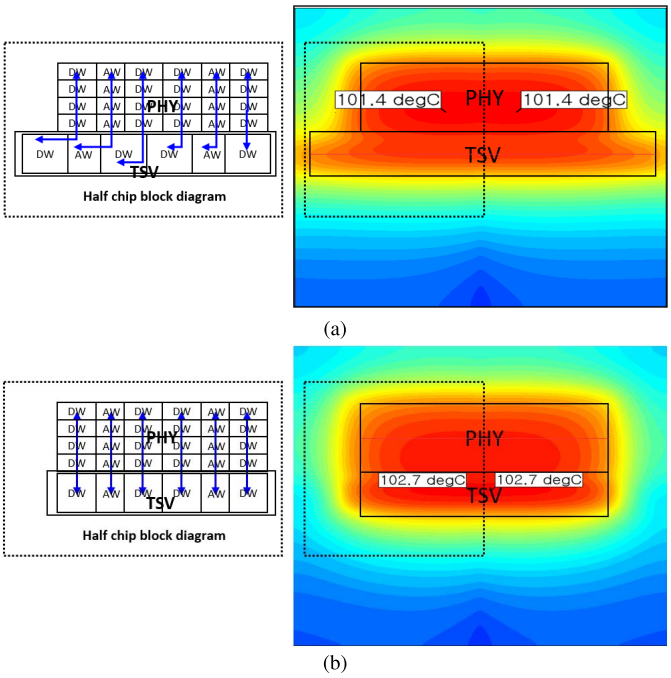


Fig. 3. Thermal analysis results of (a) TSV to core die alignment and (b) TSV to PHY alignment.

To address this issue, we exploited low-voltage signaling and machine-learning-based layout optimization that will be further explained in Section V.

III. Internal Low-Voltage Signaling Scheme

In view of the fact that HBM3 DRAM is based on vertically stacked multiple die structures with up to 12 slices, a low-power design is essential to prevent thermal issues. To manage such potential risks, a low-voltage power supply

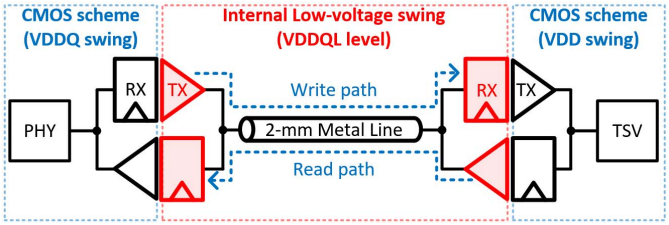


Fig. 4. Overall block diagram of the proposed internal I/O interface.

of 0.4 V, or VDDQL, is newly applied at the external I/O interface. However, thousands of internal I/Os operate with VDD or VDDQ supply, consuming a majority of the total power budget of the base die. Therefore, the VDDQL supply is also employed at the internal I/O interface, which is responsible for the PHY-to-TSV connection. With the low voltage signaling scheme, power is theoretically reduced to 86.8%, as the amount of power consumption is directly proportional to the operating voltage. Fig. 4 shows the block diagram and I/O circuits of the proposed internal I/O scheme. Bidirectional global lines toggling between VSS and VDDQL interconnect PHY and TSV, where the maximum length of the metal line is 2 mm in this design. The transmitter (Tx) and the receiver (Rx) employ an N-over-N type driver and a strong-arm latch type clocked sampler with a one-tap decision-feedback equalizer (DFE), respectively.

A. Transmitter Design and the Optimization of Global Lines

An N-over-N driver that utilizes N-type MOSFET (NMOS) for both pull-up and pull-down transistors is used as a Tx for the voltage-down conversion from VDD-level (1.1 V) to VDDQL-level (0.4 V), as shown in Fig. 5. To compensate

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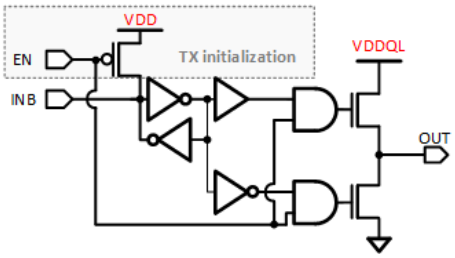
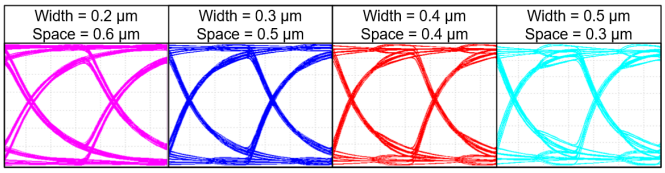
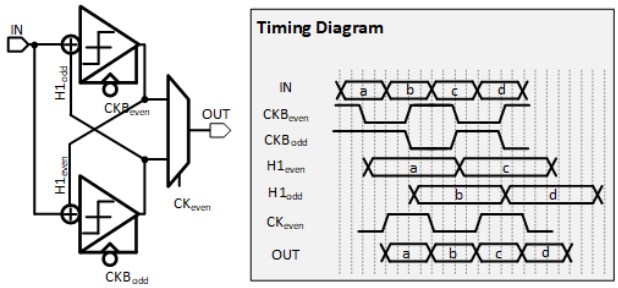


Fig. 5. Schematic of the N-over-N driver (Tx).

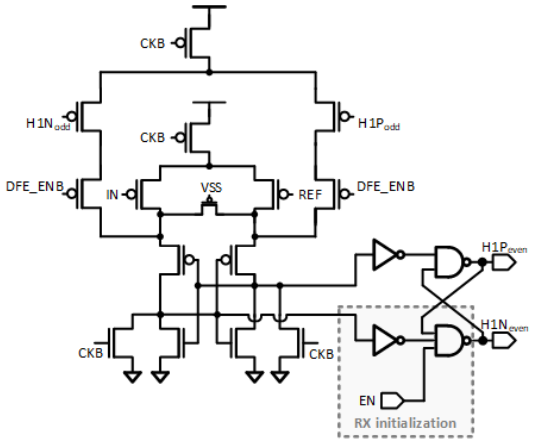


Line Pitch: 0.8μm		Deterministic Jitter (DJ)			
Width (μm)	Space (μm)	(a) GND Full Shield	(b) In-phase Toggle	(c) Out-of-phase Toggle	(c)-(a) Additional DJ Due to Xtalk
0.2	0.6	44ps	63.7ps	60.6ps	16.6ps
0.3	0.5	20.9ps	44.2ps	37.9ps	17ps
0.4	0.4	13.7ps	43.3ps	33.8ps	20.1ps
0.5	0.3	11.8ps	53.5ps	38.7ps	26.9ps

Fig. 6. Comparison table of eye diagrams and DJ over various widths and spaces of a 2-mm metal line.



(a)



(b)

Fig. 7. (a) Schematic of the clocked sense amplifier of strong-arm latch type (Rx) and its timing diagram. (b) Detailed schematic of the unit sense amplifier with DFE.

for the insufficiency of the over-drive voltage of the pull-up NMOS compared to the pull-down NMOS, the pull-up NMOS is designed with a low-Vth device.

The optimization process of global lines considering length, width, and space selection is necessary to maximize the power reduction of the low-voltage signaling scheme and to secure the sampling margin simultaneously. The line length is intentionally extended from 1.3 to 2 mm to maximize the profit of the power reduction, and the consequential inter-symbol interference (ISI) is compensated by the equalizer at the Rx side. The line pitch is limited to 0.8 μm, and line shielding for reducing crosstalk (XT) is not feasible due to the metal routing congestion. Since over 4000 internal lines need to be physically configured on the narrow TSV area, the optimization of width and space was performed within the fixed pitch of 0.8 μm. The larger width and narrower space within the defined pitch result in faster transition time due to the reduced resistance, but XT becomes larger because of bigger coupling capacitance. Fig. 6 shows the eye diagrams and the deterministic jitter (DJ) over various widths and spaces of 2-mm metal line within the fixed 0.8-μm pitch. As the DJ component due to ISI is easily compensated by the equalizer, the DJ component due to XT is difficult to predict and, therefore, difficult to be canceled out by circuit technique. As shown in the bottom table of Fig. 6, the effect of additional DJ due to XT is minimized with the linewidth and space of 0.2 and 0.6 μm. Moreover, out-of-phase toggling of adjacent lines has the advantage of the lower DJ compared to in-phase toggling. Hence, the linewidth and space are designed to 0.2 and 0.6 μm, respectively, with the out-of-phase toggling of adjacent lines.

B. Receiver Design

To recover small swing input to rail-to-rail swing, a strong-arm latch type clocked sense amplifier is employed as an Rx. A 2-mm long metal line forms a channel between Tx and Rx, with the data timing window as wide as 1tCK that corresponds to 571.4 ps for the 7-Gb/s operation. To compensate for the post-cursor ISI due to the *RC* delay caused by the physically long channel and improve voltage and timing margins, a one-tap DFE is merged to the Rx. Because the proposed I/O is a half-rate architecture, no additional latch is required for the implementation of the DFE. Fig. 7(a) shows the schematic of the proposed Rx and its timing diagram. The input signal is sampled by alternate even and odd clocks, and the decision data are fed back to the Rx reciprocally. Fig. 7(b) shows the detailed schematic of a unit sense amplifier with the DFE. A differential pair placed parallel to the input transistors adjusts the effective decision threshold level in accordance with the previous bit, followed by an SR latch that converts the strong-arm latch output from return-to-zero (RZ) to non-return-to-zero (NRZ) form. Fig. 8(a) compares the sampling margin and valid data window (tDV) with and without DFE. The corresponding sampling margins at each condition are described by the color-shaded regions. When the previous

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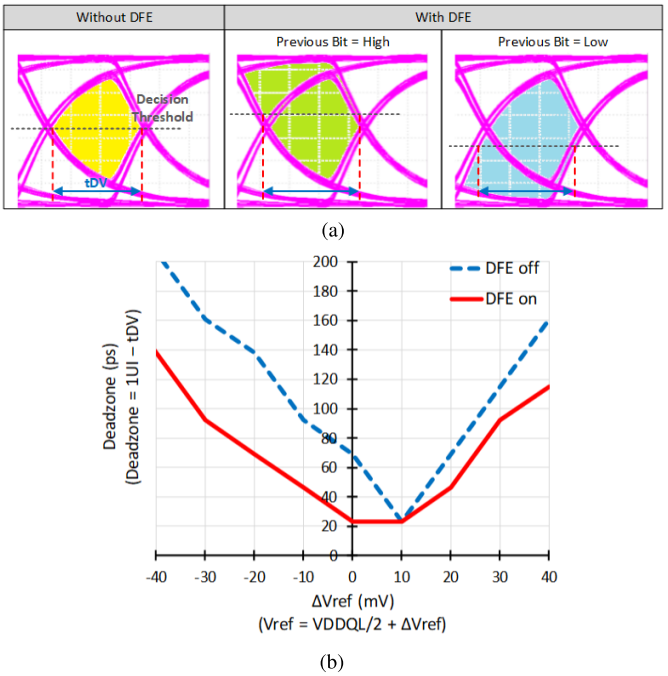


Fig. 8. (a) Sampling margin and valid data window (tDV) with and without DFE. (b) Graph of deadzone over DVref with and without DFE.

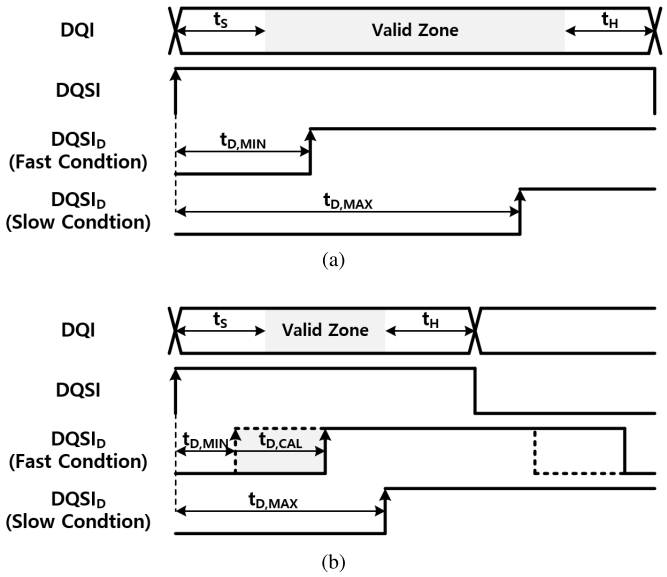


Fig. 9. (a) Previous delay match. (b) Proposed delay match.

bit is high (low), the effective decision threshold moves upward (downward) and results in a larger tDV. To verify the effect of DFE in a quantitative manner, the deadzone of the Rx is plotted in accordance with ΔVref, as shown in Fig. 8(b), where the deadzone represents the timing region where normal sampling operation fails due to setup or hold violation and ΔVref depicts the deviation of the reference voltage from the ideal reference voltage level, or half of VDDQL. The deadzone is expressed as

$$\text{Deadzone} = 1UI - tDV. \tag{1}$$

The graph shows that the DFE minimizes the deadzone, meaning that tDV becomes larger when the DFE is enabled. For the normal DFE operation, securing information of the previous bit is essential. However, during the initial state and write/read interleaving operations, there is no previous bit. Therefore, Tx and Rx latches are initialized using write/read enable signals, as highlighted by the gray-dotted box in Figs. 5 and 8(b). The Tx output is set low during initialization by the PMOS, and initial feedback data from the alternate path are set low, as well. Therefore, a proper DFE operation can be accomplished from the initial data.

By applying the internal low-voltage signaling scheme, 36% of repetitive power is reduced compared to the conventional CMOS retimer scheme that uses CMOS driver and D flip-flop type Rx under the fully toggling condition of all data lines.

IV. Center Strobe Calibration

Line skew between DQ and DQS that occurs during the transmission process through long lines is a major factor that limits operating speed. In particular, as the transmission line between PHY and TSV is bent by 90° due to the variation in their horizontal positions, the line skew increases considerably. Therefore, DQs are realigned with DQS before they are sent to the core die during the WT operation or to PHY during the RD operation to eliminate the line skew, thereby relaxing the sampling window margin on the next repeater. However, there are two problems that make the realignment difficult. First, the delay of DQS is longer than that of DQ because of the DQS distribution tree that simultaneously delivers DQS to multiple DQs. Furthermore, DQ and DQS are delivered with two different voltages, VDDQL and VDDQ, respectively. Previously, the delay line, including the clock distribution network (CDN), was designed so that the sampling edge is located within the valid window in both fast and slow operating conditions, as shown in Fig. 9(a). As the delay changes from $t_{D,\text{MIN}}$ to $t_{D,\text{MAX}}$, the valid zone must be wider than $t_{D,\text{MIN}} - t_{D,\text{MAX}}$. However, as the data rate increases, the valid window gets narrower, and it has become challengeable to obtain a sufficient sampling margin under all PVT operating conditions.

To solve this problem, this article introduces a circuit technique that automatically adjusts the strobe timing. The main idea is to adjust the offset delay so that sufficient setup and hold margin can be secured across PVT variation. As illustrated in Fig. 9(b), additional delay, $t_{D,\text{CAL}}$, is added to the default delay at the fast corner so that the total delay, $t_{D,\text{MIN}}+t_{D,\text{CAL}}$, can be bigger than the necessary setup time, t_S. On the other hand, at the slow corner, the default delay is long enough to meet the setup time constraint, and no additional delay is set to secure sufficient hold time, t_H.

Fig. 10 describes the implementation of the proposed scheme. It consists of a replica of the CDN that is twice as long as the original CDN, a delay calibration logic, six flip-flops, and an adjustable delay chain. The delay calibration logic generates DQS_CAL$\langle 0:5\rangle$ with a unit delay interval of 4t_ND2 from DQSI input, and each DQS_CAL$\langle 0:5\rangle$ samples

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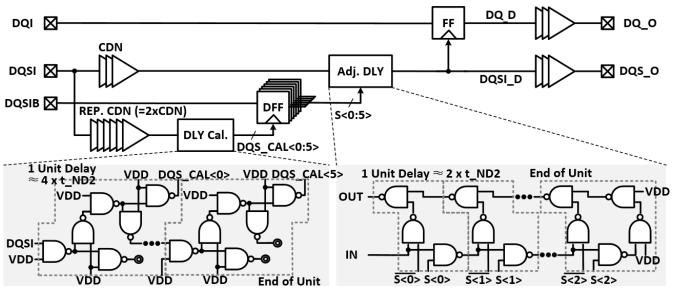


Fig. 10. Block diagram of the proposed PVT compensation center strobe scheme.

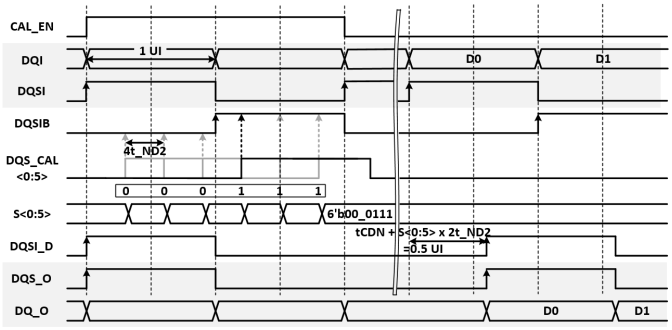


Fig. 11. Timing diagram of the center strobe scheme.

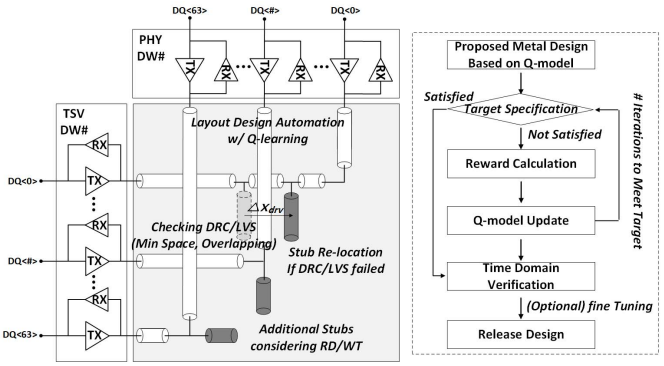


Fig. 12. Global line optimization using Q-model considering read and write operations.

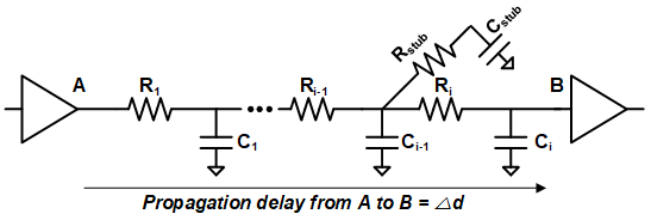


Fig. 13. Conceptual view of a global line representing a number of segmented lumped circuits to explain the Elmore delay.

DQSIB that is delayed by 1UI from the DQSI. As shown in the timing diagram in Fig. 11, the sampled values S⟨0 : 5⟩ start with "0" from the LSB, and "1" appears from a specific bit. This specific code informs the number of ND gates that make the delay of calibration logic, tDLY_{CAL}, to satisfy the following equation:

$$2\ t_{\text{CDN}} + \text{tDLY}_{\text{CAL}} = 1UI \tag{2}$$

where t_{CDN} is the delay of the original CDN. By applying this calibration code to the adjustable delay chain that has 2t_ND2 of delay unit, the setup time between DQI and its sampling clock, DQSI_D, becomes

$$1UI - (t_{\text{CDN}} + 0.5\text{tDLY}_{\text{CAL}}) = 0.5UI. \tag{3}$$

Therefore, when DQSI input passes the unit delay logic of 2tND2 by the value of S⟨0 : 5⟩, it is delayed by 0.5UI and located at the center of the DQ window. Although there is no dedicated calibration period for memory operation, the proposed circuit can be performed in a short time of 1tCK when CK is enabled.

V. Machine-Learning-Based Layout Optimization

As aforementioned, delay tuning for high-speed signals to improve the setup and hold timing margin within DWORD groups is exceptionally important, especially in wide I/O devices, such as HBM. Since the physical location of each driver and the channel parasitic of the global lines are different, it is difficult to make the signals within the same signal group have an identical amount of delay. In addition, the difficulty of group signal design would undoubtedly increase, especially when the use of vertical and horizontal global lines that have different *RC* parasitic components is essential for interconnection between PHY and TSV, or TMIO, as previously illustrated in Fig. 3. Moreover, the delay of each signal differs depending on the direction of signal transmission during read and write operations. Therefore, directionality must be considered when it comes to designing global lines targeting performance improvements. Because repetitive design changes are inevitable, it is required to find efficient design mythologies for delay tuning within the related signal groups. For design optimization of vertical and horizontal global lines, this article introduces a reinforcement-learning-based layout technique that minimizes deviation of delay of group signals, as depicted in Fig. 12.

A. Proposed Design Methodology for Global Lines

The most commonly used delay tuning method is to design an additional stub at the original line. Typically, the overall length of the channel is only taken into account to intentionally make the same value of capacitance by adding the stub. In this article, we proposed a modified equation based on the Elmore delay estimation method that considers parasitic capacitance and resistance to improve delay estimation accuracy.

A propagation delay of a signal can be simply calculated considering an additional loading, as depicted in Fig. 13, as follows:

$$\Delta d = ln(\beta)\left(\sum_{i=1}^{n} C_i \sum_{j=1}^{i} R_j + C_{\text{stub}} \sum_{k=1}^{k=i-1} R_k\right) \tag{4}$$

where $ln(\beta)$ is typically 0.69 in theory and valid at the ideal input signal, but the equation must be slightly modified based

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on the actual design. Consequently, the optimal global line design for each driver could be effectively determined in terms of the length and the location of an additional stub as follows:

$$\max\{\Delta d_{\forall i}\} - \Delta d_i \cong \Delta d_i + \alpha_i \tag{5}$$

$$\alpha_i \cong ln(\beta) \cdot \sum_{k=1}^{i-1} R_i C_{M\#_\text{unit}} L_{\text{stub}} \tag{6}$$

where Δd_i and α_i refer to a propagation delay and the required additional delay at the driver i, respectively. In addition, a coefficient β is a specific value that can be extracted from SPICE simulations. The coefficient varies depending on the semiconductor process and the driver strength. C_M#_unit stands for the capacitance per unit length of a certain metal selected for a stub design. An ideal design for each signal can be calculated in relation to an additional parasitic capacitance due to a stub. Especially, the length of a stub for each signal is precisely estimated by targeting the slowest driver within target signal groups.

From the previous equation, the optimum position and optimum length can be simultaneously estimated for a number of target signals. However, the location where the stub can be designed is limited due to the significant number of other global lines adjacent to the signals. Therefore, the additional stubs can be overlapped with other global lines and result in an impractical solution. To solve this problem, we propose a Q-model to find the closest solution, that is, the best possible solution. The Q-model can successfully achieve the goals while learning based on the following rewards in the equation:

$$\text{Reward} = R_{\text{learn}} \cdot \left(\sum_{i=1}^{\#\text{of signals}} \Delta x_{\text{drv}} \right)^{-1} \tag{7}$$

where R_{learn} is the coefficient for a learning rate to iterate the Q-model. Δx_{drv} is the distance from the previous stub position when it has to move to another position due to a short between other metals. To secure a workable layout design to achieve meaningful results in terms of group signal skew, the Q-model repeatedly conducts the given reward calculation and updates the Q-model based on the reward. Empirically, it is possible to obtain comparatively reasonable results after performing reinforcement learning for approximately 100 000 times.

B. Simulation-Based Verification for Worst DWORD Group

To verify the proposed design algorithm for delay tuning, eye-diagram simulations are conducted considering read and write operations with the worst channel, where the channel length is the longest between PHY and TSV. Based on the simulations, it is possible to clearly evaluate the signal integrity of group signals by overlapping all DWORD data. Signal integrity of a design applying a machined-learning-based layout optimization algorithm is compared with the original design using the conventional capacitance matching method, as summarized in Table II and depicted in Fig. 14, respectively. It is clear that the proposed design methodology shows improvements in timing margin of up to a maximum of 33% at read operation at slow PVT conditions even with reduced time loss for iterations between a schematic designer and a layout designer.

TABLE II
SUMMARY OF AN EFFECTIVE GROUP SIGNAL SKEW

Operation	DWORD	PVT	Original(ps)	Proposed(ns)
Write	Group 1	Fast	99.2	106
		Slow	132.0	107.0
	Group 2	Fast	76.5	57.8
		Slow	92.6	77.6
Read	Group 1	Fast	71.8	53.3
		Slow	103.0	69.2
	Group 2	Fast	68.3	46.9
		Slow	91.4	73.8

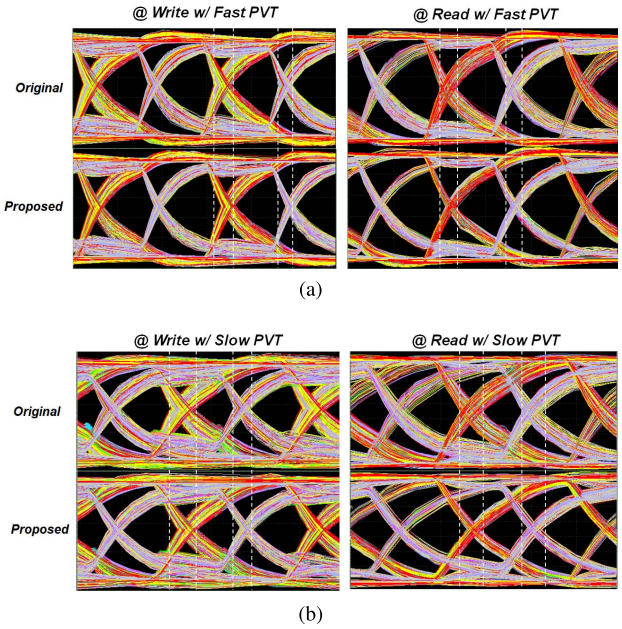


Fig. 14. Eye-diagram result and comparison with a previous design with (a) fast and (b) slow PVTs at write and read operations.

VI. TSV CALIBRATION

The TSV interface is another key bottleneck for achieving high data rates. In general, HBM uses point-to-point signal transmission (between each core die and base die), which reduces signal loads and is insensitive to process skew variations. However, as higher density is required from HBM devices without major architectural changes, the multiple slices are connected to ranks (SIDs) among the eight-high stack structure. The vertical eight-high memory stack consists of multiple ranks that share I/O ports among TSVs on the same channel. When core slices with different process skew corners are jointly stacked, the overall speed degrades due to a lack of tCCDR (column-to-column access between ranks) timing, which derives from the CLK-to-output delay (tDQSCK of core die) difference between each slice during the rank-to-rank read operation.

Although the conventional TSV calibration reduces tCCDR timing, it cannot be implemented if the available TSV I/O area is limited due to the complexity related to the long operating

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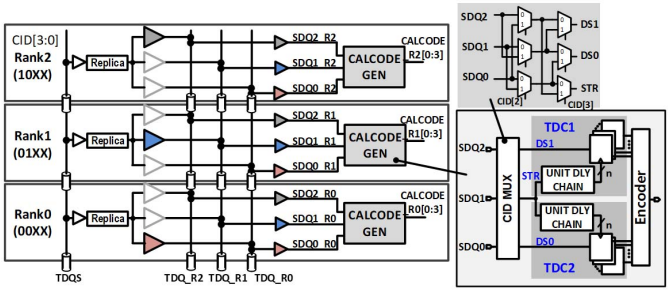


Fig. 15. TSV rank-to-rank timing calibration circuit.

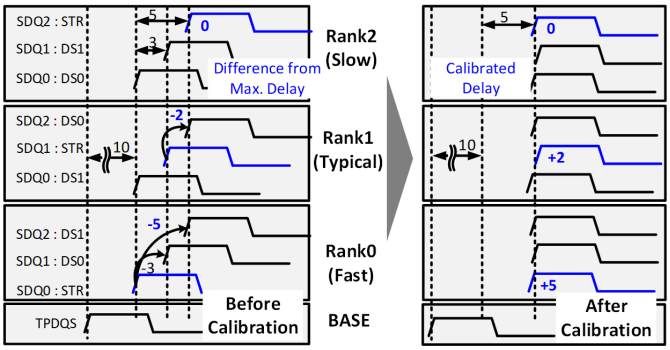


Fig. 16. TSV rank-to-rank timing calibration operation timing.

period, especially when more than three ranks are configured [13]. Another scheme presented for 3-D DRAM [14] also has a short calibration time, but there are limitations because the TSV load of each slice is indirectly compared with TSV modeling. Fig. 15 shows the proposed TSV rank-to-rank calibration circuit physically fit into the limited area with a short calibration time. For independent normal read operations, it uses a dedicated control signal called TDQS, which sends the replicated delay through TSV (TDQ_R0-R2) to a delay skew detector circuit located in each slice called CALCODE GEN. CID MUX in CALCODE GEN uses physical slice ID (CID) to distinguish the current slice strobe signal (STR) and the object signals (DS0, DS1) for time-to-digital converter (TDC) comparison. TDC calculates each skew difference among multi-ranks, simultaneously, and stores the result code in the register, sending the encoded difference value to all TSV I/O circuit regions.

The TSV rank-to-rank calibration operation can be exemplified by the change in delay between before and after the calibration. Rank 0, 1, and 2 are classified as fast, typical, and slow skews, respectively, in Fig. 16. It is assumed that the relative differences in delay per skew for high-speed operation are +3 and +5, respectively. The delay difference is detected by TDC, and each delay is compensated by digital codes; the skew difference between slices after delay calibration is satisfied within the TDC unit delay. Fig. 17 shows the measurement result of the TSV calibration. The analog code is the actual delay gap between two slices expressed in the ring oscillator delay unit, and it can be clearly seen that the analog code difference between Slices 4 and 0 well matches

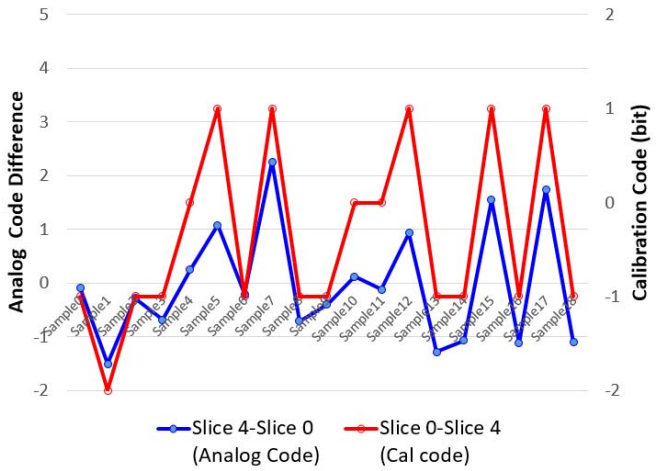


Fig. 17. Comparison of the measured delay difference and the calibration code.

the digital calibration code difference between slices after TSV calibration.

VII. On-Die ECC Based on Reed–Solomon Code

To remove DRAM malfunctions originated from cell failures, a symbol-based OD ECC is adapted. Fig. 18 shows the cell array organization and the proposed ECC scheme. To recover both multi- and single-bit errors, a (19, 17) single symbol error correction Reed–Solomon code was implemented in the proposed OD ECC engine. Since malfunction by a sub-word line (SWL) is one of the most fatal error cases, each symbol is set to 16 bits coming from an SWL. In total, there are a total of 17 symbols, including 16 symbols of normal DQs and one symbol of metadata used for system-level RAS feature implementation. The proposed ECC scheme needs two symbols of parity bits, which are generated from the other 17 symbols during the write operation and stored in separate ECC cells. The two parity symbols and 17 data symbols are used for the generation of 32-bit syndrome during the read operation, where the last 16 bits represent the failed symbol location, and the starting 16 bits represent the failed bits in the symbol. When an error occurs, any error in a symbol can be corrected using the syndrome. The proposed scheme provides not only the corrected data but also the real-time severity information in four categories—no error (NE), correctable error single-bit error (CEs), correctable error multi-bit error (CEm), and uncorrectable error (UE). Furthermore, as HBM3 DRAM provides an additional one symbol of metadata, system-level RAS features, such as ECC or CRC, can be implemented to further improve the reliability.

A. Error Correction & Detection Coverage of HBM3 ECC Engine

To check the correction and detection capability of the proposed ECC scheme, we tested the implemented circuit with the most frequently observed errors, such as the multi-bit errors caused by the malfunction of the SWL or SWL driver (SWD), and single-bit or dual-bit intermittent errors. Fig. 19

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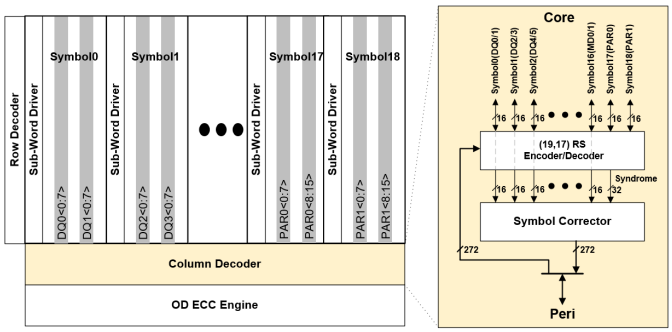


Fig. 18. OD ECC scheme based on the (19, 17) Reed–Solomon code and its correction capability.

Fault mode / ECC operation	1 SWL fault	1 SWD fault	Intermittent error	
			Single bit	Double bit
Correction	100 %	0 %	100 %	4.9505 %
Miscorrection	0 %	0.0015 %	0 %	9.3994 %
Detection	0 %	99.9985 %	0 %	85.6501 %

Fig. 19. Proposed ECC correction and detection capability.

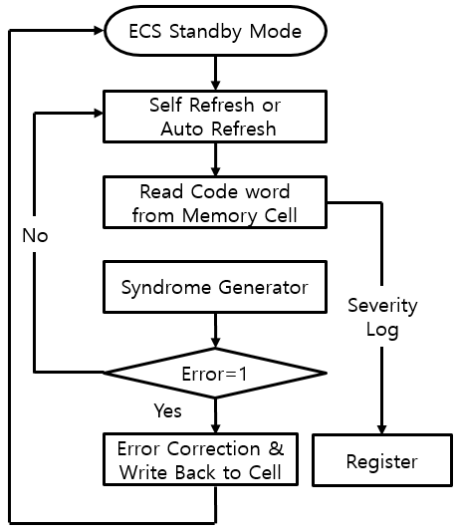


Fig. 20. State diagram of the ECS operation.

summarizes the test results. As aforementioned, one SWL fault is perfectly corrected regardless of the number of erroneous bits because a symbol represents an SWL. When an SWD malfunctions, errors occur in two symbols for 0.0015% cases, but 99.9985% of the errors are still detected where 4 294 836 225 cases are possible calculated as

$$\#\text{ of error cases in 2 symbols} - \#\text{ of 1 symbol errors} = \left(2^{32}-1\right) - \left(2^{16}-1\right) \times 2 = 4294836225. \quad (8)$$

In the case of intermittent 1-bit errors, since it is an error within one symbol, 100% correction is possible, but, in the case of 2-bit errors, 4.9505 % can be corrected, and 85.6501 % can be detected, but 9.3994 % among ${}_{304}C_2$ cannot be corrected or detected. It is possible to further improve the capability, but the focus of the implementation is the operating speed and minimized area. Therefore, we implemented an ECC engine that supports 896-GB/s operation with 100% fatal SWL error correction capability.

TABLE III
KEY FEATURES' SUMMARY

	HBM3 (This Work)
Pin Data Rate	7Gbps
Max Bandwidth	896GB/s
Max Density	16Gb x 12-High = 192Gb
Burst Length	8
Organization	16 Channel x 2 Pseudo-Channel x 32 I/O
Supply Voltage	VDDC/Q=1.1V, VDDQL=0.4V, VPPE=1.8V
Die Dimension	11mm×11mm

B. Error Check and Scrub Scheme

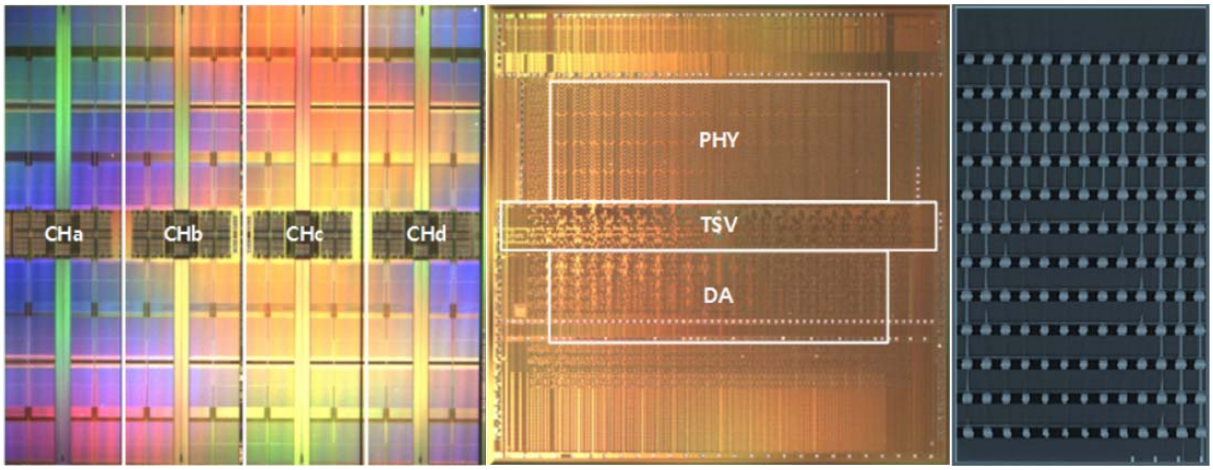
In order to improve the reliability of DRAM, the error check and scrub (ECS) function is provided for each pseudo-channel. As shown in Fig. 20, ECS operates automatically during the refresh operation, records the error log to a register accessible through the IEEE 1500 interface, and removes intermittent errors at the same time. The log includes the error type—NE, correctable single-bit error, correctable multi-bit error, and UE—and the address of bank, row, and column address. The error log is beneficial to understand the severity of errors that occur in the entire DRAM cells, and the data correction prevents the accumulation of intermittent errors from becoming a UE.

VIII. EXPERIMENTAL RESULTS

A 16-Gb HBM3 DRAM chip is implemented in a 1z-nm DRAM process, and up to 12 core dies are stacked on a base die and connected via TSVs. Fig. 21(a) and (b) shows the chip photos of the fabricated HBM3 core die and base die, respectively. In addition, the cross-sectional images of the 12-Hi stacked structure are shown in Fig. 21(c). The package size is 11 × 11 mm in line with industry consensus, while the package height is 28.3 mils or 720 μm. The thickness of the core dies, excluding the top and bottommost dies, is approximately about 30 um, which is 40% thinner than the prior products.

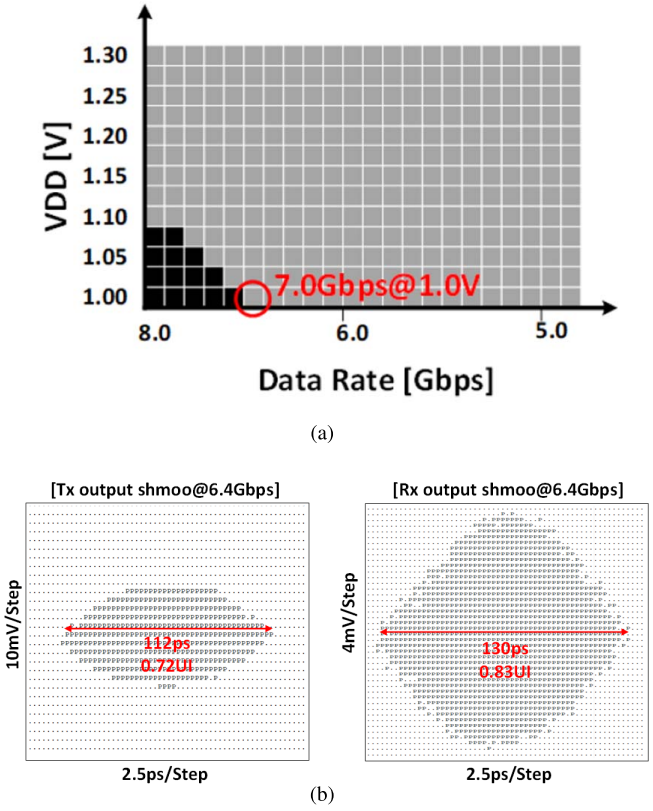
The overall performance is shown in Fig. 22(a) that illustrates the Shmoo plots for tCK and VDD. It achieves 7.0 Gb/s per pin at 1.0-V supply voltage, which is equivalent to 857 GB/s of the total bandwidth. At 1.067 V, which is the minimum operating voltage specified by the JEDEC HBM3 DRAM spec, it operates at 7.4 Gb/s per pin. The performance measurement was done with a passive interposer package (PIP) fabricated with the DRAM process. As the

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(a) (b) (c)

Fig. 21. (a) DRAM die. (b) Base die logic. (c) 12-high TSV stacks.



(a)

(b)

Fig. 22. (a) tCK versus VDD Shmoo. (b) Tx and Rx IO Shmoo.

HBM PHY interface is not accessible before being integrated into a System in Package (SiP), an in-house interposer is necessary for the ATE test. The measured I/O test results of the Shmoo plot are shown in Fig. 22(b). The measured eye openings are 112 ps (0.72UI) and 120 mV for Tx and 130 ps (0.83UI) and 160 mV for Rx at 6.4 Gb/s/pin, respectively.

Table III summarizes the key features of this work. It supports more density with faster operating speed and lower power compared to the previous HBM2E. The total density is 192 Gb, but the physical density is 228 Gb as 12-high stacked HBM includes 24 Gb of parity cells for in-DRAM ECC and an additional 12 Gb of metadata cells. The total bandwidth is improved by 75%, and power efficiency has been improved by up to 15% due to its lower operating voltage where the supply voltages for VDD/VDDQ/VDDQL/VPP are 1.1/1.1/0.4/1.8 V, respectively.

IX. Conclusion

This article presented a 192-Gb 12-high stacked HBM3 DRAM that supports max 896-GB/s bandwidth. We described various circuit techniques that enable high data rate operation with low power consumption. The center strobe calibration scheme successfully improved the operating speed by adjusting the strobe timing across PVT variations, and signal skews caused by variation of signal lines between PHY and TSV were effectively minimized with the proposed machine-learning-based layout optimization technique. The timing conflict at TSV nodes caused by the difference in CLK-to-out delay among core die slices was resolved by the automatic TSV calibration scheme. The high-speed operation was achieved with lower power consumption compared to previous work mainly due to the lower supply voltage and internal low-voltage signaling scheme. This work also improved the reliability of the DRAM by implementing a symbol-correcting OD ECC scheme based on the Reed–Solomon code. All in all, the proposed HBM3 DRAM is expected to accommodate increasing system-level demands, including higher memory bandwidth, density, power efficiency, and reliability.

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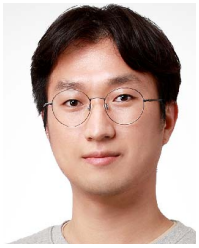
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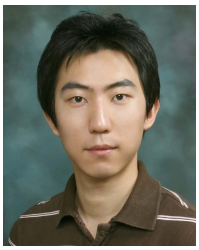
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